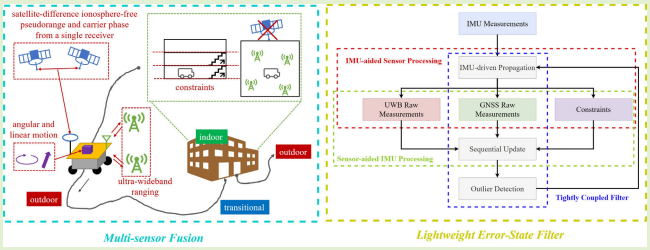
An Indoor and Outdoor Seamless Positioning System for Low-Cost UGV Using PPP/INS/UWB Tightly Coupled Integration

Xingxing Li, Zongzhou Wu, Zhiheng Shen, Zhili Xu, Xin Li, Shengyu Li, and Junjie Han

***Abstract*—Seamless positioning is critical for mobile robots to achieve ubiquitous availability in all scenarios. The precise point positioning (PPP) technique provides a centimeter-level solution without a reference station and is often integrated with an inertial navigation system (INS) for outdoor navigation of unmanned ground vehicles (UGVs). However, in challenging environments such as indoors, the performance of PPP/INS systems is severely degraded, while ultrawideband (UWB) is widely used due to its high accuracy over short distances. This article proposes a tightly coupled (TC) PPP/INS/UWB integrated system, in which satellite-difference ionosphere-free (SD-IF) pseudorange and carrier-phase measurements from a single receiver, microelectromechanical system (MEMS) inertial measurements, and the UWB rangings are tightly fused to accomplish continuous and precise positioning throughout indoor and outdoor coverage. To mitigate the effects of nonline-of-sight (NLOS), multipath, and faulty signals, a two-step weighting approach is proposed to improve UWB positioning performance, including segmentation weighting based on the signal strength, and weight adjustment based on area identification. Besides, motion constraints are imposed on velocity, yaw, and height to suppress drifts during signal-harsh periods. Real-world experiment results show that the proposed system can achieve continuous positioning in the overall indoor and outdoor environments, with the mean absolute errors (MAEs) of (0.214, 0.186, and 0.243 m) and the root mean square errors (RMSEs) of (0.339, 0.291, and 0.414 m) in the east, north, and up directions. In transitional areas with severe signal interference and interruption, a 0.5-m positioning accuracy and smooth switching are still attainable.**



***Index Terms*— Area identification, microelectromechanical system (MEMS), multisensor fusion, precise point positioning (PPP), seamless positioning, tightly coupled (TC) integration, ultrawideband (UWB).**

I. Introduction

UBIQUITOUS localization is a prerequisite for cross-domain location-based services (LBSs) [1], [2]. The National Human Activity Pattern Survey (NHAPS) indicated that approximately 87% of the time is spent indoors, while only 7%–8% is spent outdoors [3]. Hence, a challenge for unmanned ground vehicles (UGVs) serving human activities is to achieve seamless positioning throughout outdoor [4] and indoor [5], [6] coverage.

For outdoor environments, the global navigation satellite system (GNSS) provides all-weather, all-day, high-precision positioning, navigation, and timing (PNT) services to users worldwide [7], [8], [9]. Among the GNSS positioning methods, precise point positioning (PPP) [10] is favored in recent years because it provides a global solution without the requirement for a local reference station, and has a flexible range. However, limited by transmission power and path attenuation, the performance of GNSS positioning represented by PPP can be significantly degraded when subjected to interference (e.g., urban canyons, tunnels, and intersections [2]). An inertial navigation system (INS) is a high-bandwidth autonomous system that provides accurate relative attitude, velocity, and position estimation [11], and is considered to be quite effective for short-term GNSS outages [8], [12]. In [13], the PPP/INS system was proposed and verified for the first time. The literature [14] compared the performance of the loosely coupled (LC) and tightly coupled (TC) integration, indicating that the latter has a more stable performance in GNSS-harsh environments. In [15] and [16], the carrier-phase ambiguity resolution was applied to the PPP/INS system to reduce the convergence time of float solutions. The literature [17], [18], [19], [20] has proved the PPP/INS system can be enhanced with motion

Manuscript received 12 June 2023; revised 14 August 2023; accepted 28 August 2023. Date of publication 6 September 2023; date of current version 16 October 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB2501102, in part by the National Natural Science Foundation of China under Grant 41974027, and in part by the Sino-German Mobility Program under Grant M-0054. The associate editor coordinating the review of this article and approving it for publication was Dr. Geethu Joseph. *(Corresponding author: Zhiheng Shen.)*

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Digital Object Identifier 10.1109/JSEN.2023.3310480

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constraints and additional sensors. In [21], factor graph optimization (FGO) was used to further improve performance through multiple iterations and full utilization of historical information. In the open sky, PPP/INS integration is capable of obtaining centimeter-level positioning [17]. But for long-term outages in GNSS-denied environments (e.g., indoors), the navigation error accumulates in the integrated system, especially for those incorporating low-cost microelectromechanical system (MEMS) inertial measurement units (IMUs).

To reproduce outdoor precise positioning indoors, ultrawideband (UWB), a technology similar to GNSS, is considered a good substitute [22]. The technology is a carrier-free communication that uses nanosecond to picosecond nonsinusoidal narrow pulses to transmit extremely low-power signals over a wide spectrum [23]. Due to high precision and moderate cost, UWB has been applied to smart terminals (e.g., iPhone 11 series and Xiaomi MIX 4) in recent years. Typically, UWB works in anchor-tag [24] or network ranging [25] mode. The schemes based on received signal strength indication (RSSI), time of arrival (TOA), time difference of arrival (TDOA), and angle of arrival (AOA) have been applied to UWB positioning [26]. Nevertheless, UWB can only achieve communication over tens of meters [27], which is not suitable for larger areas. When reflections are severe, UWB ranging degrades due to inherent susceptibility to interference. In UWB-based indoor positioning, the nonline-of-sight (NLOS) identification and multipath mitigation have attracted attention [28], [29], [30]. However, in the field of indoor and outdoor seamless positioning involving multiple areas and larger scenarios, the effects of NLOS, multipath, and faulty signals are more significant and still deserve further exploration.

For the extremely challenging task of seamless indoor and outdoor positioning, several novel sensor fusion approaches have been proposed in recent years. In [31], light detection and ranging (LiDAR) was loosely integrated into a real-time kinematic (RTK)/INS system for uninterrupted positioning and 2-D mapping indoors. In [32], a TC RTK/INS/UWB/Map integration system with map-aided NLOS detection was proposed, and the authors mentioned that the system performed well for tactical-grade IMU but not for MEMS-IMU. The two methods use measurements from a local station, which limits the application scale. In [33], a TC method based on GNSS carrier-phase, tactical-grade IMU, and UWB was proposed and evaluated at preset static points. The method required a large number of UWB anchors to ensure no rank deficit. In [34], a PPP/INS/UWB TC filter was proposed to enhance the performance of PPP/INS. This literature only illustrated the importance of UWB weighting outdoors but did not focus on indoor-outdoor continuous positioning. From the studies above, seamless positioning systems have not been fully investigated, especially on continuous switching of multiple scenarios, mitigation of faulty signals, and effects of sensor degradation, which are the main purposes of our work. For practical considerations, our research is based on low-cost industrial-grade sensors, rather than expensive ones.

In this article, a TC PPP/INS/UWB indoor-outdoor seamless positioning system for low-cost UGVs is proposed. The state propagation is driven by IMU precise mechanization, while measurements that come from a single GNSS receiver, UWB, and motion constraints are employed to correct IMU navigation drifts. Raw GNSS pseudorange and carrier-phase measurements are used in a satellite-difference ionosphere-free (SD-IF) way to avoid the effects of ionosphere delay and low-cost receiver clock bias. A two-step weighting model for UWB range measurements is proposed to attenuate the ranging accuracy degradation caused by NLOS, multipath, and faulty signals. The operation illustration of the proposed system is shown in Fig. 1. The main contributions of this article are as follows.

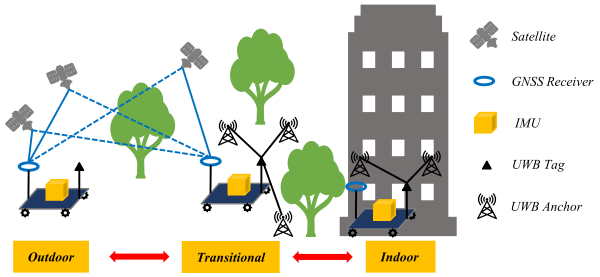


Fig. 1. Simplified operation illustration of the proposed seamless positioning system.

1) A TC PPP/INS/UWB seamless positioning system is proposed, in which a MEMS IMU is integrated with SD-IF pseudorange and carrier-phase from a single GNSS receiver, motion constraints, and UWB ranging. In addition, area identification and cross-area switching can be achieved without images and maps.
2) A two-step weighting model including RSSI-based segmentation weighting and area-based weight adjustment is proposed to suppress outliers in UWB rangings. Benefiting from this, our method can suppress outliers in heterogeneous sensor fusion to minimize degradation caused by area switching.
3) The performance of the proposed system is extensively evaluated by real-world experiments covering indoor and outdoor scenarios. Several aspects not fully addressed by existing work (e.g., switching between scenes, degradation resistance, and efficiency) are validated in detail.

The rest of this article is organized as follows. Sensor models and motion constraints are presented in Section II. The TC integration algorithm and an overview of the proposed system are shown in Section III. The experimental results are analyzed in Section IV. Finally, Section V presents our conclusions and prospects for future work.

II. Methodology

Several coordinate frames and transformations are involved in this section. For clarity, the main frames in this article are defined as follows.

1) *e-Frame:* Earth-centered Earth-fixed (ECEF) frame.
2) *b-Frame:* IMU body frame with a right-front-up axis.
3) *n-Frame:* Local frame with an east-north-up axis.
4) *i-Frame:* Ideal inertial frame.

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A. Inertial Navigation Model

The motion can be described by a set of continuous-time differential equations. After the initial states are given, the attitude, velocity, and position in the next step are calculated through numerical integration, which is known as mechanization. In inertial navigation for e-frame [35], these differential equations are written as

$$\begin{bmatrix} \dot{\boldsymbol{C}}_b^e \\ \dot{\boldsymbol{v}}^e \\ \dot{\boldsymbol{r}}^e \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_b^e(\boldsymbol{\omega}_{eb}^b\times) \\ \boldsymbol{C}_b^e\boldsymbol{f}_{ib}^b - 2\boldsymbol{\omega}_{ie}^e \times \boldsymbol{v}^e + \boldsymbol{g}^e \\ \boldsymbol{v}^e \end{bmatrix} \tag{1}$$

where $\boldsymbol{C}_b^e$ denotes the direction cosine matrix (DCM) from b-frame to e-frame. $\boldsymbol{\omega}_{eb}^b$ denotes the rotation angular rate vector of b-frame with respect to e-frame in b-frame. $\boldsymbol{\omega}_{ie}^e$ is the Earth rotation angular rate vector projected to e-frame. $(\cdot)\times$ is the skew-symmetric operator of vectors. $\boldsymbol{f}_{ib}^b$ denotes the specific force in b-frame. $\boldsymbol{g}^e$ denotes gravity vector in e-frame. $\boldsymbol{v}^e$ and $\boldsymbol{r}^e$ denote velocity and position vectors in e-frame.

The propagation of state errors, also known as the error model, describes the time-varying characteristics of attitude, velocity, and position errors [36]. Perform error perturbation on (1), and the continuous-time model is written as

$$\begin{bmatrix} \delta\dot{\boldsymbol{\alpha}}_b^e \\ \delta\dot{\boldsymbol{v}}^e \\ \delta\dot{\boldsymbol{r}}^e \end{bmatrix} = \begin{bmatrix} -(\boldsymbol{\omega}_{ie}^e\times)\delta\boldsymbol{\alpha}_b^e - \boldsymbol{C}_b^e\delta\boldsymbol{\omega}_{ib}^b \\ \delta\boldsymbol{g}^e + (\boldsymbol{C}_b^e\boldsymbol{f}_{ib}^b)\times\delta\boldsymbol{\alpha}_b^e - 2(\boldsymbol{\omega}_{ie}^e\times)\delta\boldsymbol{v}^e + \boldsymbol{C}_b^e\delta\boldsymbol{f}_{ib}^b \\ \delta\boldsymbol{v}^e \end{bmatrix} \tag{2}$$

where δ denotes the error operator for vectors in (1). $\delta\boldsymbol{\alpha}_b^e$ denotes the misalignment angle vector in e-frame. $\delta\boldsymbol{\omega}_{ib}^b$ and $\delta\boldsymbol{f}_{ib}^b$ denote the measurement errors of the gyroscope and accelerometer.

B. Satellite-Difference Ionosphere-Free PPP Model

The ionosphere is a complex dispersive medium that has a significant impact on GNSS measurements. The ionosphere-free (IF) model is often used in PPP to eliminate the first-order term of the ionosphere delay [10], which is written as

$$\begin{cases} P_{\mathrm{IF}} = \rho + c\cdot(\delta t_r - \delta t^s) + m\cdot(T_h + T_w) + \varepsilon_{P,\mathrm{IF}} \\ \varphi_{\mathrm{IF}} = \rho + c\cdot(\delta t_r - \delta t^s) + m\cdot(T_h + T_w) + \lambda_{\mathrm{IF}}\cdot N_{\mathrm{IF}} + \varepsilon_{\varphi,\mathrm{IF}} \end{cases} \tag{3}$$

where P_{IF} and φ_{IF} denote the IF pseudorange and carrier-phase measurements. ρ denotes the geometric distance between the satellite and the receiver. c denotes the speed of light. δt_r and δt^s denote the receiver-side and satellite-side clock bias. m denotes the tropospheric mapping function. T_h and T_w denote the zenith hydrostatic and wet troposphere delay. λ_{IF} and N_{IF} denote the wavelength and ambiguity corresponding to the IF carrier-phase combination. Among them, the carrier-phase ambiguity is kept constant over successive satellite tracking. $\varepsilon_{P,\mathrm{IF}}$ and $\varepsilon_{\varphi,\mathrm{IF}}$ denote the observation noise and other unmodeled errors on IF pseudorange and phase measurements.

In general, satellites with higher elevation angles have less chance of interference. Therefore, the satellite elevation angle el can be used to weight observations [37]. For the pseudorange and carrier-phase measurements, the standard deviations of $\varepsilon_{P,\mathrm{IF}}$ and $\varepsilon_{\varphi,\mathrm{IF}}$ can be expressed as

$$\begin{cases} \sigma_{P,\mathrm{IF}}^{\mathrm{sys}} = \kappa\cdot\sigma_P^{\mathrm{sys}}\cdot[2\sin(\mathrm{el})]^{-1} \\ \sigma_{\varphi,\mathrm{IF}}^{\mathrm{sys}} = \kappa\cdot\sigma_\varphi^{\mathrm{sys}}\cdot[2\sin(\mathrm{el})]^{-1} \\ \kappa = \sqrt{(\lambda_1^2+\lambda_2^2)/(\lambda_1^2-\lambda_2^2)} \end{cases} \tag{4}$$

where σ_P^{sys} and $\sigma_\varphi^{\mathrm{sys}}$ denote the pseudorange and carrier-phase prior standard deviations of satellite system sys. κ is the coefficient introduced by combination and can be derived from the coefficient of the IF model. λ_j denotes the GNSS-carrier wavelength of jth frequency.

On the one hand, the hardware delays of different satellite system signals are different, resulting in additional time biases [38] that need to be considered in the model. On the other hand, the clock biases of low-cost receivers are unstable and generally require no less than four satellites to accurately solve. Therefore, in our work, a more simplified SD-IF PPP model is adopted, written as

$$\begin{cases} \nabla P_{\mathrm{IF}} = \nabla\rho - \nabla\delta t^s + \nabla m\cdot(T_h + T_w) + \varepsilon_{\nabla P,\mathrm{IF}} \\ \nabla\varphi_{\mathrm{IF}} = \nabla\rho - \nabla\delta t^s + \nabla m\cdot(T_h + T_w) + \lambda_{\mathrm{IF}}\cdot\nabla N_{\mathrm{IF}} + \varepsilon_{\nabla\varphi,\mathrm{IF}} \end{cases} \tag{5}$$

where ∇ denotes the satellite-difference operator. T_h can be corrected by model [39], and δt^s can be corrected by precise products. It is worth noting that SD will lead to correlation among satellites due to the same reference, which needs to be specified by matrix $\boldsymbol{H}_{\mathrm{SD}}$. There is only one 1 and one -1 in each row of the matrix $\boldsymbol{H}_{\mathrm{SD}}$ [40], and the measurement variance matrix after SD can be written as

$$\boldsymbol{R}_{\nabla\mathrm{IF}} = \boldsymbol{H}_{\mathrm{SD}}\cdot\boldsymbol{R}_{\mathrm{IF}}\cdot\boldsymbol{H}_{\mathrm{SD}}^T \tag{6}$$

where $\boldsymbol{R}_{\mathrm{IF}}$ denotes the measurement variance matrix of the IF model calculated according to (4).

C. UWB Positioning Model Based on Two-Step Weighting

UWB can obtain the rangings between tag and anchors to achieve positioning. The simplified mathematical model between tag and anchor i is

$$\begin{aligned} d_i &= \left\|\boldsymbol{r}_t^e - \boldsymbol{r}_{a,i}^e\right\| + \varepsilon_{d,i} \\ &= \sqrt{(x_t - x_{a,i})^2 + (y_t - y_{a,i})^2 + (z_t - z_{a,i})^2} + \varepsilon_{d,i} \end{aligned} \tag{7}$$

where d_i denotes the measured distance. $\|\cdot\|$ denotes the Euclidean norm of vectors. $\boldsymbol{r}_t = (x_t, y_t, z_t)^T$ denotes the coordinate of the tag. $\boldsymbol{r}_{a,i} = (x_{a,i}, y_{a,i}, z_{a,i})^T$ denotes the coordinate of the anchor i. $\varepsilon_{d,i}$ denotes the measurement noise.

With the rangings with multiple precalibrated anchors, the position of the user tag can be captured by the intersection principle. However, severe NLOS, multipath, and faulty signals significantly reduce the UWB positioning performance, especially in complex indoor scenarios. Also, different ranging signals are inconsistently affected by multipath and NLOS due to the differences in the path and signal strength. Generally, the RSSI can reflect the signal power drop caused by path attenuation and obstructions [41]. Unlike stochastic

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modeling procedures that use equal weight [33] or require the calculation of measurement residuals [37], the RSSI-based model implements stochastic modeling directly using signal strength, which is more consistent with actual measurement precision [41] and avoids the impact of estimator performance. Hence, we propose a two-step weighting model for UWB rangings to attenuate the degradation caused by NLOS, multipath, and faulty signals, including segmentation weighting based on RSSI and weight adjustment based on area identification.

1) Segmentation Weighting Based on RSSI: Due to NLOS and multipath, the received signals are often not exactly from the first path. Considering that nonfirst path signals can also be captured, we utilize the RSSIs of the total received signals to determine the rough weights of measurements. Further, the multipath and NLOS detection, and segmentation weighting are realized by the RSSI difference between the total received and the first path signals. Referring to the carrier-to-noise ratio weighting method [42], the standard deviations of the rangings can be written as

$$\sigma_d = \beta \cdot \sigma_0 \cdot \sqrt{10^{\frac{-\mathrm{RSSI_{tr}}}{10}}} \tag{8}$$

$$\beta = \begin{cases} 1, & 0\ \mathrm{dBm} < \mathrm{RSSI_{tr}} - \mathrm{RSSI_{fp}} \le 10\ \mathrm{dBm} \\ 3, & 10\ \mathrm{dBm} < \mathrm{RSSI_{tr}} - \mathrm{RSSI_{fp}} \le 20\ \mathrm{dBm} \\ \infty, & \mathrm{RSSI_{tr}} - \mathrm{RSSI_{fp}} > 20\ \mathrm{dBm} \end{cases} \tag{9}$$

where σ_0 denotes the rough ranging error, which can be found in the device manual. $\mathrm{RSSI_{tr}}$ and $\mathrm{RSSI_{fp}}$ denote the RSSIs of the total received signal and the first path signal, respectively. β denotes the weight factor for low-quality rangings.

2) Weight Adjustment Based on Area Identification: Given the ability of UWB signals to penetrate walls [43], the user tag may receive signals with large gross errors (called faulty signals) from anchors located in other areas, which would seriously damage ranging performance. Therefore, it is necessary to zone and reject these faulty signals to achieve robust positioning. The UWB anchors that are precisely located in advance can assist in identifying the current area to exclude faulty signals. At the same time, the closer the anchor to the tag, the lower the probability of being affected by NLOS and multipath, which is also worth considering to adjust the segmentation RSSI weighting.

Different from NLOS identification methods through images [44], [45] or prior maps [32], [34], in our work, according to the relative position between the tag and the anchors, area identification, faulty signal deletion, and weight adjustment are realized. Anchors are arranged outdoors and indoors, and the areas where the tag is located are divided into indoor, outdoor, and transitional types. Following the steps of Algorithm 1, we find four special points, i.e., the outdoor nearest anchor (abbreviated as ON), the outdoor reference center (abbreviated as OR), the indoor nearest anchor (abbreviated as IN), and the indoor reference center (abbreviated as IR). The nearest anchor is defined by Euclidean distance. The reference point is considered to be the center of the anchors within the area. When the distribution of anchors in the area is uneven, the area can be divided into multiple blocks for processing. Then, we calculate $\boldsymbol{u}_{\mathrm{OR}}$, $\boldsymbol{u}_{\mathrm{ON}}$, $\boldsymbol{u}_{\mathrm{IR}}$, $\boldsymbol{u}_{\mathrm{IN}}$, which are the direction cosine vectors from the tag to the four points in n-frame, as shown in Fig. 2.

Algorithm 1 Area-Based Weight Adjustment

Input: $\mathcal{F}$=($\mathrm{Area}_{t,k-1}$, $\boldsymbol{r}_t^e$, $\mathcal{A}_i$=$\{\boldsymbol{r}_{a,i}^e$, $\sigma_{d,i}$,$\mathrm{Area}_{a,i}\}$).
Output: $\mathcal{S}$=($\mathrm{Area}_{t,k}$, $\mathcal{V}_i$=$\{\sigma_{d,i}\}$).

$\mathrm{Area}_{t,k} = \mathrm{Area}_{t,k-1}$;
Calculate: $\boldsymbol{u}_{sp}$=$\boldsymbol{C}_e^n(\boldsymbol{r}_{sp}^e$-$\boldsymbol{r}_t^e)$, sp=ON, OR, IN, IR;
if $\frac{\|\boldsymbol{u}_{IR}\|}{\|\boldsymbol{u}_{OR}\|} > \zeta$ **or** $\boldsymbol{u}_{OR} \cdot \boldsymbol{u}_{ON} < 0$ **then**
  $\mathrm{Area}_{t,k}$ = outdoor;
else if $\frac{\|\boldsymbol{u}_{OR}\|}{\|\boldsymbol{u}_{IR}\|} > \zeta$ **or** $\boldsymbol{u}_{IR} \cdot \boldsymbol{u}_{IN} < 0$ **then**
  $\mathrm{Area}_{t,k}$ = indoor;
else if $\boldsymbol{u}_{OR} \cdot \boldsymbol{u}_{IR} < 0$ **then**
  $\mathrm{Area}_{t,k}$ = transitional;
end
for $i \in \mathcal{F}$ **do**
  Calculate: $d_{a,i} = \|\boldsymbol{r}_t^e - \boldsymbol{r}_{a,i}^e\|$;
  if $\mathrm{Area}_{t,k} \neq$ transitional *and* $\mathrm{Area}_{a,i} \neq \mathrm{Area}_{t,k}$ **then**
    Delete i in $\mathcal{F}$; **continue**;
  end
  if $\mathrm{Area}_{a,i}$ = indoor **then**
    $f_i = \frac{d_{a,i}}{\|\boldsymbol{r}_{IR}^e - \boldsymbol{r}_{a,i}^e\|}$;
  else if $\mathrm{Area}_{a,i}$ = outdoor **then**
    $f_i = \frac{d_{a,i}}{\|\boldsymbol{r}_{OR}^e - \boldsymbol{r}_{a,i}^e\|}$;
  end
  $\sigma_{d,i} = f_i \cdot \sigma_{d,i}$; Add $\sigma_{d,i}$ to $\mathcal{V}_i$;
end
Add ($\mathrm{Area}_{t,k}$, $\mathcal{V}_i$) to $\mathcal{S}$.

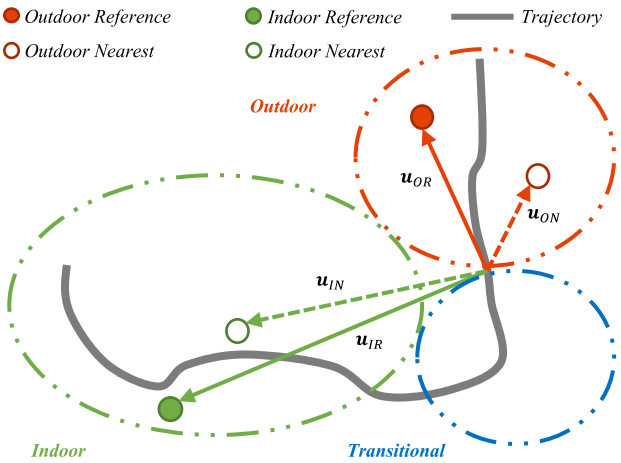


Fig. 2. Schematic of anchor-based area identification algorithm.

For step k, the algorithm input $\mathcal{F}$ includes the tag area $\mathrm{Area}_{t,k-1}$ of step $k-1$, the current tag position $\boldsymbol{r}_t^e$, and the set $\mathcal{A}_i$, which includes the position, ranging standard deviation, and area of each anchor. First, set the threshold ζ, and through the relative position ratio, we can determine the current area $\mathrm{Area}_{t,k}$ of the tag. Then, the anchors outside the area are deleted, and the weight adjustment factor f_i is determined by the distance between the tag and each anchor. Next, the factor is employed to modify the measurement standard deviation $\sigma_{d,i}$ input by the algorithm. Finally, the $\mathrm{Area}_{t,k}$ and standard

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deviations of all anchors $\mathcal{V}_i$ are added to algorithm output $\mathcal{S}$, while the input-related variables are updated to the next step of algorithm execution.

D. Motion Constraints

Motion constraints can enhance system performance without additional hardware costs and are ideal for low-cost applications [20]. Positioning in an open sky can be guaranteed by GNSS, so challenging environments are mainly shaded and indoor areas. Considering that UGV alternately appears stationary and moving, and the height is unchanged on the same floor indoors, we utilize zero velocity update (ZUPT), yaw constraint, and height constraint to enhance the system performance in these environments.

1) Zero Velocity Update: When stationary relative to the ground, the ZUPT can be used to construct a constraint. The equation can be written as

$$\boldsymbol{v}_{\mathrm{INS}}^{e} - \mathbf{0}_{3\times 3} = \delta \boldsymbol{v}^{e} + \boldsymbol{\varepsilon}_{\mathrm{ZUPT}} \tag{10}$$

where $\boldsymbol{v}_{\mathrm{INS}}^{e}$ denotes the velocity in e-frame obtained by the INS model. $\boldsymbol{\varepsilon}_{\mathrm{ZUPT}}$ denotes the measurement noise.

2) Yaw Constraint: Due to the weak observability of the vertical gyroscope bias, it is difficult for low-cost MEMS IMUs to obtain accurate yaw angles, which could lead to rapid drifts of both the yaw angle error and position error in the signal-harsh environments. To improve, referring to the dual-antenna attitude constraint in [46], the equation of the yaw constraint can be written as

$$\psi - \psi_{\mathrm{INS}} = \boldsymbol{H}_{\psi} \delta \boldsymbol{\alpha}_{b}^{e} + \varepsilon_{\psi} \tag{11}$$

where ψ and ψ_{INS} denote the final yaw before standstill and the yaw obtained by the INS model. ε_{ψ} denotes the measurement noise. The specific form of $\boldsymbol{H}_{\psi}$ can be found in [46].

3) Height Constraint: When the UGV moves on the same floor without overturning and tipping, the height remains the same. The height can come from calculations or be measured by external methods. Perturb the conversion between e-frame coordinate and latitude B, longitude L, and height h, the height constraint equation can be written as in (12)–(14), shown at the bottom of the page.

Here h denotes the height of the IMU center. h_{INS} denotes the height obtained by the INS model. $\boldsymbol{H}_h$ denotes the Jacobian matrix. $\boldsymbol{H}_{h,3}$ represents the third row of $\boldsymbol{H}_h$. R_P denotes the Earth's prime vertical radius of curvature. w denotes the derivative of R_p with respect to latitude B. e denotes the Earth's eccentricity. ε_h denotes the measurement noise.

III. Tightly Coupled Integration

A. Filter Prediction

According to Section II, in addition to the parameters of the INS model, there are the zenith wet troposphere delay and carrier-phase ambiguity parameters of the SD-IF PPP model, while the parameters of UWB and motion constraints can be incorporated into the INS model. Hence, the state vector of the integrated system consists of two parts, i.e., INS states and GNSS states, written as

$$\boldsymbol{X} = \left[\boldsymbol{X}_{\mathrm{INS}}^{T}, \boldsymbol{X}_{\mathrm{GNSS}}^{T}\right]^{T} \tag{15}$$

$$\boldsymbol{X}_{\mathrm{INS}} = \left[\left(\delta \boldsymbol{\alpha}_{b}^{e}\right)^{T}, \left(\delta \boldsymbol{v}^{e}\right)^{T}, \left(\delta \boldsymbol{r}^{e}\right)^{T}, \left(\delta \boldsymbol{b}_{g}\right)^{T}, \left(\delta \boldsymbol{b}_{a}\right)^{T}\right]^{T} \tag{16}$$

$$\boldsymbol{X}_{\mathrm{GNSS}} = \left[T_w, \left(\nabla \boldsymbol{N}_{\mathrm{IF}}\right)^{T}\right]^{T} \tag{17}$$

where $\delta \boldsymbol{b}_g$ and $\delta \boldsymbol{b}_a$ denote the gyroscope and accelerometer bias error vectors. $\nabla \boldsymbol{N}_{\mathrm{IF}}$ denotes the vector form of all ∇N_{IF} SD pairs. To unify the reference points of different sensors, we convert the position errors of GNSS $\delta \boldsymbol{r}_{\mathrm{GNSS}}$ and UWB $\delta \boldsymbol{r}_{\mathrm{UWB}}$ to IMU position error $\delta \boldsymbol{r}^{e}$ based on extrinsic perturbation as

$$\delta \boldsymbol{r}_{\mathrm{GNSS}} = -\left[\delta \boldsymbol{r}^{e} + \left(\boldsymbol{C}_{b}^{e} \boldsymbol{l}_{\mathrm{GNSS}}^{b}\right) \times \delta \boldsymbol{\alpha}_{b}^{e}\right] \tag{18}$$

$$\delta \boldsymbol{r}_{\mathrm{UWB}} = -\left[\delta \boldsymbol{r}^{e} + \left(\boldsymbol{C}_{b}^{e} \boldsymbol{l}_{\mathrm{UWB}}^{b}\right) \times \delta \boldsymbol{\alpha}_{b}^{e}\right] \tag{19}$$

where $\boldsymbol{l}_{\mathrm{GNSS}}^{b}$ and $\boldsymbol{l}_{\mathrm{UWB}}^{b}$ are the lever arm vectors from the IMU center to GNSS and UWB center in b-frame, respectively.

In INS states, the $\delta \boldsymbol{\alpha}_{b}^{e}$ uncertainty comes from the noise of angle random walk (ARW). The $\delta \boldsymbol{v}^{e}$ uncertainty comes from the noise of velocity random walk (VRW). The $\delta \boldsymbol{b}_g$ and $\delta \boldsymbol{b}_a$ are modeled as random walk processes. In GNSS states, T_w is modeled as a random walk process, and ∇N_{IF} is modeled as a random constant process. The filter prediction model can be written as

$$\dot{\boldsymbol{X}} = \boldsymbol{A}\boldsymbol{X} + \boldsymbol{W} \tag{20}$$

$$\boldsymbol{A} = \left[\begin{array}{ccccc|cc}
-\boldsymbol{\omega}_{ie}^{e}\times & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & -\boldsymbol{C}_{b}^{e} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times j} \\
\left(\boldsymbol{C}_{b}^{e}\boldsymbol{f}_{ib}^{b}\right)\times & -2\boldsymbol{\omega}_{ie}^{e}\times & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \boldsymbol{C}_{b}^{e} & \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times j} \\
\mathbf{0}_{3\times 3} & \boldsymbol{I}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times j} \\
\mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times j} \\
\mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 1} & \mathbf{0}_{3\times j} \\
\hline
\mathbf{0}_{1\times 3} & \mathbf{0}_{1\times 3} & \mathbf{0}_{1\times 3} & \mathbf{0}_{1\times 3} & \mathbf{0}_{1\times 3} & 0_{1\times 1} & \mathbf{0}_{1\times j} \\
\mathbf{0}_{j\times 3} & \mathbf{0}_{j\times 3} & \mathbf{0}_{j\times 3} & \mathbf{0}_{j\times 3} & \mathbf{0}_{j\times 3} & \mathbf{0}_{j\times 1} & \mathbf{0}_{j\times j}
\end{array}\right] \tag{21}$$

where $\boldsymbol{W}$ is the vector of process noise. $\boldsymbol{I}_{n\times n}$ denotes the n-dimension unit matrix. The covariance matrix of process

$$h_{\mathrm{INS}} - h = \boldsymbol{H}_{h,3} \delta \boldsymbol{r}^{e} + \varepsilon_h. \tag{12}$$

$$\boldsymbol{H}_h = \left[\begin{array}{ccc}
w\cos B\cos L - (R_P + h)\sin B\cos L & -(R_P + h)\cos B\sin L & \cos B\cos L \\
w\cos B\sin L - (R_P + h)\sin B\sin L & (R_P + h)\cos B\cos L & \cos B\sin L \\
w\sin B + [R_P(1 - e^2) + h]\cos B & 0 & \sin B
\end{array}\right]^{-1} \tag{13}$$

$$w = \frac{dR_P}{dB} = \frac{d\left(a\Big/\sqrt{1 - e^2\sin^2 B}\right)}{dB} = \frac{ae^2\sin B\cos B}{\sqrt{\left(1 - e^2\sin^2 B\right)^3}} \tag{14}$$

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noise $\boldsymbol{Q}_{\boldsymbol{W}}$ is

$$\begin{aligned}\boldsymbol{Q}_{\boldsymbol{W}} &= \mathrm{diag}(\boldsymbol{Q}_{\mathrm{INS}}, \boldsymbol{Q}_{\mathrm{GNSS}})\\ \boldsymbol{Q}_{\mathrm{INS}} &= \mathrm{diag}\big(P_{\mathrm{ARW}}^2\boldsymbol{I}_{3\times3}, P_{\mathrm{VRW}}^2\boldsymbol{I}_{3\times3},\\ &\qquad P_r^2\boldsymbol{I}_{3\times3}, P_{b_g}^2\boldsymbol{I}_{3\times3}, P_{b_a}^2\boldsymbol{I}_{3\times3}\big)\\ \boldsymbol{Q}_{\mathrm{GNSS}} &= \mathrm{diag}\big(P_{T_w}^2, P_{\nabla N_{\mathrm{IF}}}^2\boldsymbol{I}_{j\times j}\big)\end{aligned} \tag{22}$$

where diag(·) denotes the block diagonal matrix. P denotes the power spectral density of the process noise. j denotes the number of SD pairs.

B. Filter Update

Suppose the measurements are available at the current step k, which may include parts of GNSS, UWB, and the motion constraints. After linearization, the filter update is as

$$\boldsymbol{Z}_k = \boldsymbol{H}\boldsymbol{X}_k + \boldsymbol{V}_k \tag{23}$$

$$\boldsymbol{Z}_k = \begin{bmatrix} \nabla\boldsymbol{P}_{\mathrm{IF}} - \nabla\boldsymbol{\rho}_{\mathrm{INS}} \\ \nabla\boldsymbol{\varphi}_{\mathrm{IF}} - \nabla\boldsymbol{\rho}_{\mathrm{INS}} \\ \boldsymbol{d} - \boldsymbol{d}_{\mathrm{INS}} \\ \boldsymbol{v}_{\mathrm{INS}}^e - \boldsymbol{0} \\ \psi - \psi_{\mathrm{INS}} \\ h_{\mathrm{INS}} - h \end{bmatrix} \tag{24}$$

$$\boldsymbol{H} = \left[\begin{array}{cccccc|cc} \nabla\boldsymbol{g}\cdot(\boldsymbol{C}_b^e\boldsymbol{l}_{\mathrm{GNSS}}^b)\times & \boldsymbol{0}_{j\times3} & \nabla\boldsymbol{g} & \boldsymbol{0}_{j\times3} & \boldsymbol{0}_{j\times3} & \nabla\boldsymbol{m} & \boldsymbol{0}_{j\times j} \\ \nabla\boldsymbol{g}\cdot(\boldsymbol{C}_b^e\boldsymbol{l}_{\mathrm{GNSS}}^b)\times & \boldsymbol{0}_{j\times3} & \nabla\boldsymbol{g} & \boldsymbol{0}_{j\times3} & \boldsymbol{0}_{j\times3} & \nabla\boldsymbol{m} & \lambda_{\mathrm{IF}}\boldsymbol{I}_{j\times j} \\ \boldsymbol{u}\cdot(\boldsymbol{C}_b^e\boldsymbol{l}_{\mathrm{UWB}}^b)\times & \boldsymbol{0}_{n\times3} & \boldsymbol{u} & \boldsymbol{0}_{n\times3} & \boldsymbol{0}_{n\times3} & \boldsymbol{0}_{n\times1} & \boldsymbol{0}_{n\times j} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times1} & \boldsymbol{0}_{3\times j} \\ \boldsymbol{H}_{\psi} & \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & 0_{1\times1} & \boldsymbol{0}_{1\times j} \\ \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & \boldsymbol{H}_{h,3} & \boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & 0_{1\times1} & \boldsymbol{0}_{1\times j} \end{array}\right] \tag{25}$$

$$\boldsymbol{\rho}_{\mathrm{INS}} = \left\|\boldsymbol{r}^e + \boldsymbol{C}_b^e\boldsymbol{l}_{\mathrm{GNSS}}^b - \boldsymbol{r}^s\right\| \tag{26}$$

$$\boldsymbol{d}_{\mathrm{INS}} = \left\|\boldsymbol{r}^e + \boldsymbol{C}_b^e\boldsymbol{l}_{\mathrm{UWB}}^b - \boldsymbol{r}_{a,i}^e\right\| \tag{27}$$

where $\boldsymbol{V}_k$ is the vector of measurement noise at step k. $\boldsymbol{g}$ denotes the unit direction cosine vectors from the receiver to all satellites. $\boldsymbol{u}$ denotes the unit direction cosine vectors from the tag to all anchors. n denotes the number of available anchors. $\boldsymbol{\rho}_{\mathrm{INS}}$ and $\boldsymbol{d}_{\mathrm{INS}}$ denote the zero-order terms in the linearization computed by inertial propagation. $\boldsymbol{r}^s$ denotes the position vector of an observed satellite. The measurement variance matrix $\boldsymbol{R}_{\boldsymbol{V}}$ is a block diagonal matrix

$$\boldsymbol{R}_{\boldsymbol{V}} = \mathrm{diag}(\boldsymbol{R}_{\nabla\mathrm{IF}}, \boldsymbol{R}_{\mathrm{UWB}}, \boldsymbol{R}_{\mathrm{ZUPT}}, \boldsymbol{R}_{\psi}, \boldsymbol{R}_h). \tag{28}$$

Because of the different sample rates and acquisition conditions of various measurements, for the step with missing measurements, the matrix is a subpart of the above form.

C. System Overview

The indoor-outdoor seamless positioning system proposed in this article is based on a lightweight extended Kalman filter (EKF). From the perspective of filter design, the scheme fuses GNSS measurements, i.e., pseudorange and carrier-phase from a single receiver, IMU gyroscope and accelerometer records, and UWB rangings at raw measurement level, i.e., TC integration. In addition, motion constraints are also introduced into the filter to enhance performance.

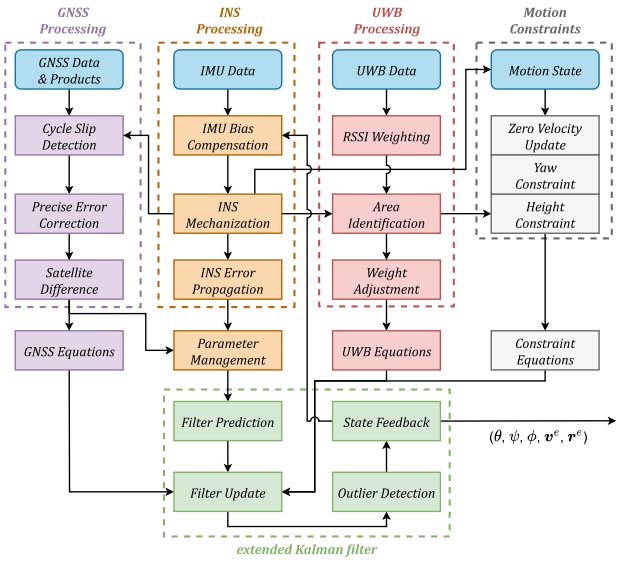


Fig. 3. Architecture of the proposed TC indoor–outdoor seamless positioning system.

The architecture of the multisensor fusion system is depicted in Fig. 3. The filter is initialized by GNSS measurements outdoors. Next, the relative estimations from the high-rate inertial mechanization are used to assist in the preprocessing of heterogeneous sensors, i.e., cycle slip detection of GNSS carrier-phase, anchor-based area identification, and motion state recognition. After the preprocessing of these sensors, measurements by outlier detection are used to aid in correcting IMU error propagation to benefit from IMU autonomy for continuous indoor-outdoor positioning and orientation.

The lightweight EKF we use only stores the error states at the current step, which easily meets the efficiency requirements of real-time applications. Given the different sampling rates of multisource measurements, we employ a sequential strategy that progressively processes each type of measurement. On the one hand, it is convenient for the filter to perform outlier detection and robustness processing of the same measurements. On the other hand, it avoids the problem of a sharp dimension increase in the measurement matrix. After each filter update, the method in [47] is used to recursively detect, identify, and adapt (DIA) outliers in real-time.

IV. Experimental Results

A. Experiment Description

To evaluate the performance of the proposed seamless positioning system, a real-world experiment was carried out on the campus of Wuhan University, as given in Fig. 4. The experiment included not only outdoor open sky, one side shaded, and heavily shaded scenarios, but also indoor corridor and room scenarios. A low-cost sensor suite was used, including an MEMS ADIS-16470 IMU, a u-blox ZED-F9P receiver connected to a Multi-GNSS HT-3701B antenna, and a Nooploop LinkTrack P-B UWB tag. Three outdoor anchors were erected under the trees to aid positioning during

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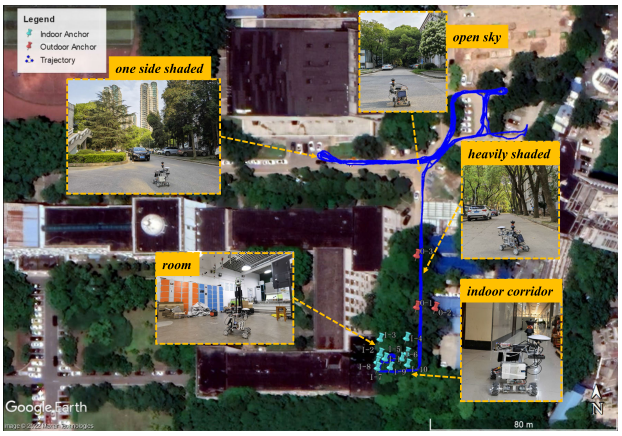


Fig. 4. Top view of experimental trajectory and images of typical scenes (red markers: outdoor anchors; cyan markers: indoor anchors).

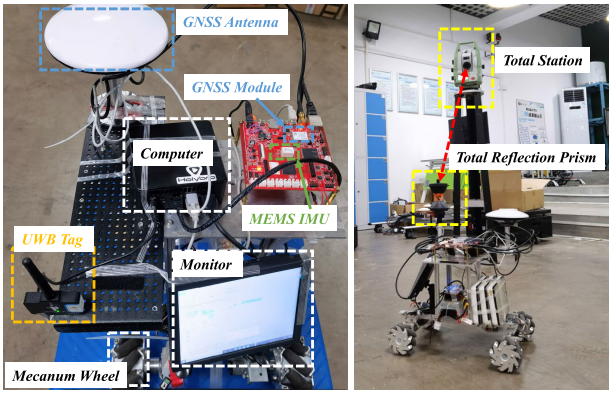


Fig. 5. Experimental vehicle equipment (left: sensor suite; right: equipment for automatic tracking).

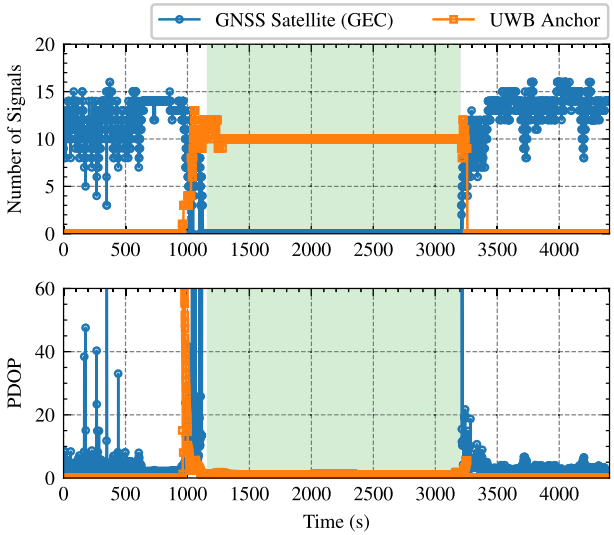


Fig. 6. Number of signals (top) and the PDOP (bottom) in the real-world vehicle test. The green background color indicates indoor data periods.

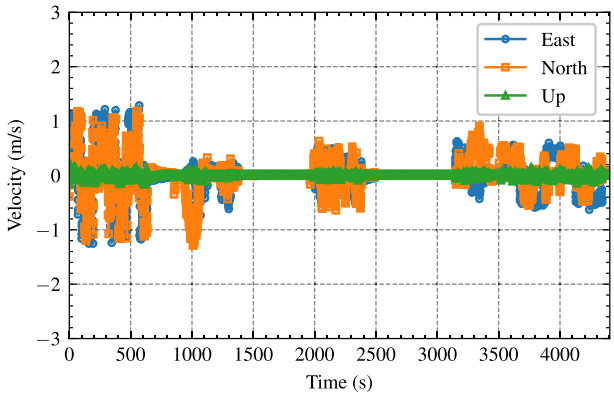


Fig. 7. Velocity time series in the experiment.

GNSS degradation, and the remaining ten indoor anchors were attached to corridors and room walls to ensure continuous positioning indoors. The sampling rates of the IMU, GNSS, and UWB were 100, 1, and 10 Hz, respectively. During the experiment, an additional total reflection prism was fixed to facilitate optical tracking of the total station. The appearance of the suite is shown in Fig. 5. For PPP processing, the rapid precise orbit and clock products provided by the German Research Centre for Geosciences (GFZ) were used.

The experimental datasets were collected by an Intel NUC mini computer (Intel Core i7-1165G7) fixed on the remote control vehicle, simulating a UGV moving back and forth indoors and outdoors. Fig. 6 shows the changes in the signal number and position dilution of precision (PDOP), where GNSS satellites include GPS (abbreviated as G), Galileo (abbreviated as E), and BDS (abbreviated as C). In the experiment, the vehicle switched between indoor and outdoor environments twice, and there were different GNSS visibilities outdoors. Consequently, the experiment was considered to be a scenario-diverse and challenging evaluation condition. In shaded and indoor areas, a Leica MS-50 total station set on known locations was used to measure and obtain a nominal millimeter-level reference trajectory. In open areas, we verified that the ambiguity resolution rate of Multi-GNSS RTK exceeds 90%, so that outside the MS-50 tracking range, the reference trajectory was the Multi-GNSS RTK/INS TC solution (the distance between the local station and the vehicle is not more than 1 km). Fig. 7 shows the velocity sequence in the vehicle experiment. From 1550 to 1950 s and 2650 to 3050 s, the vehicle remains stationary for accuracy evaluation on static points.

B. Positioning and Orientation Performance

In this section, we explore and compare the performance of the three fusion algorithms in detail. P^{3}I denotes the TC integration of SD-IF PPP and INS. P^{3}IU denotes the TC integration of SD-IF PPP, INS, and equal-weight UWB, which is seen as a conventional method. P^{3}IUC denotes the proposed TC integration method.

The positioning error sequences in the east, north, and up directions throughout the experiment are shown in Fig. 8. Before 900 s, only GNSS and IMU measurements are available for all three methods to have identical positioning errors. Thereafter, until 3200 s, the dense shade causes a significant number reduction of GNSS satellites and there is a complete break for GNSS indoors, so the P^{3}I method rapidly diverges due to the lack of sufficient measurements. Thanks to the UWB rangings, both the P^{3}IU method and the proposed method can achieve continuous positioning in the heavily

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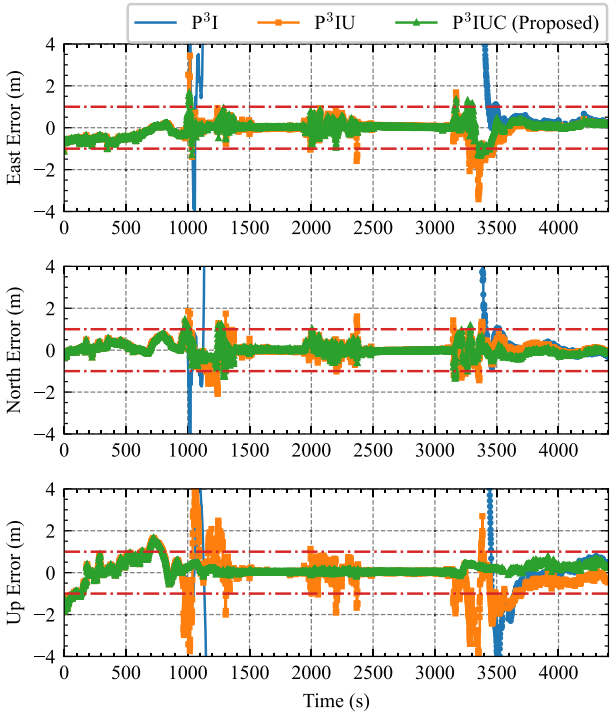


Fig. 8. Positioning error sequences in east, north, and up directions (red dotted line: ± 1 m).

TABLE I
POSITIONING PERFORMANCE OF DIFFERENT METHODS

Method	MAE (m)			RMSE (m)			Availability*
	East	North	Up	East	North	Up	
P³I	0.571	0.344	1.109	1.041	0.796	2.528	31.82%
P³IU	0.248	0.221	0.513	0.416	0.354	0.865	79.64%
Proposed	**0.214**	**0.186**	**0.243**	**0.339**	**0.291**	**0.414**	**90.67%**
Improved(%)**	13.71	15.84	**52.63**	18.51	17.80	**52.14**	**11.03**

* The horizontal and vertical errors are both less than 1m.
** From P³IU to the proposed method.

GNSS-degraded area and indoor area. However, the conventional P³IU method shows meter-level solutions during indoor-outdoor switching and indoor motion periods, especially in the up direction. Since UWB is the dominant measurement during this period, we believe that these outliers stem from severe NLOS, multipath and faulty signals caused by indoor walls and facilities. In contrast, the proposed method mitigates the interference of NLOS, multipath and faulty signals through the two-step weighting model and multiple motion constraints to significantly improve the performance during the period. After 3200 s, the P³I and P³IU methods exhibit different magnitudes of reconvergence, while the proposed method achieves smoother switching. The main reason is that when GNSS signals recover, the values and variances of the GNSS carrier-phase ambiguity parameters are reinitialized, which indirectly increases the position uncertainty. The proposed method improves the position estimation through more accurate prior information provided by the height constraint, which will be further indicated later.

The mean absolute errors (MAEs), root mean square errors (RMSEs), and availabilities are further compared in Table I. The MAEs in the east, north, and up directions are improved from (0.248, 0.221, and 0.513 m) of the P³IU method to (0.214, 0.186, and 0.243 m) of the proposed method. The RMSEs in the east, north, and up directions are improved from (0.416, 0.354, and 0.865 m) of the P³IU method to (0.339, 0.291, and 0.414 m) of the proposed method. Under the definition that both horizontal and vertical errors are less than 1m, the positioning availability is improved from 79.64% of the P³IU method to 90.67% of our method.

In order to reveal the roles of UWB rangings and motion constraints for indoor-outdoor seamless positioning and orientation performance, the standard deviations of the attitude, velocity, and position error states are given in Fig. 9, respectively. Due to the lack of adequate measurements, the standard deviations of P³I diverge indoors. Compared with the P³I method, the P³IU method suppresses the drift of attitude, velocity, and position through UWB measurements. The ZUPT and yaw constraint represented by the red background can significantly improve the attitude and velocity estimations in the proposed method. As mentioned above, the GNSS carrier-phase ambiguity parameters are reinitialized after reentering outdoors, which increases the position uncertainties of the P³I and P³IU methods to the meter level. In this reconvergence stage, the ZUPT and yaw constraint are not triggered due to the continuous motion during area switching, while the height constraint significantly reduces position uncertainty, which corresponds to the results in Fig. 8. In conclusion, our method has direct motion constraints on the attitude, velocity, and position parameters to ensure stable and reliable indoor-outdoor pose estimation.

C. Accuracies in Typical Scenarios

Considering the area types in this article, as well as static periods in indoor environments, the accuracies of four typical periods including outdoor, transitional, indoor, and static points are separately counted for detailed evaluation. The boxplots of 3-D positioning errors are given in Fig. 10. The median 3-D errors of the conventional P³IU method are (0.764, 0.096, and 0.061 m) for outdoor, indoor, and static points, while that of the proposed method are (0.604, 0.072, and 0.041 m), with (20.94%, 25.00%, 32.79%) improvements, respectively. Moreover, it is worth noting that in the transitional area, the median 3-D error of the P³IU method increases sharply to 1.412 m, making it difficult to smoothly interface indoor and outdoor positioning. In contrast, the median 3-D error of our method is 0.496 m with 64.87% improvement, enabling more stable switching. In four typical scenarios, the errors of the proposed method are significantly lower than the P³IU method.

The cumulative distribution functions (cdfs) of the 3-D positioning errors for typical periods are given in Fig. 11. Due to the convergence of carrier phase ambiguity, most of the outdoor 3-D errors are at decimeter level. In the indoor area, UWB measurements can provide positioning from centimeters to decimeters, and further, centimeter-level positioning can be achieved at static points. However, in the transitional area, the proportion of the conventional P³IU method with 3-D errors less than 1 and 2 m is only 40.19% and 64.24%, respectively, which are significant degradations compared to both indoor and outdoor areas. In contrast, our method achieves 93.04%

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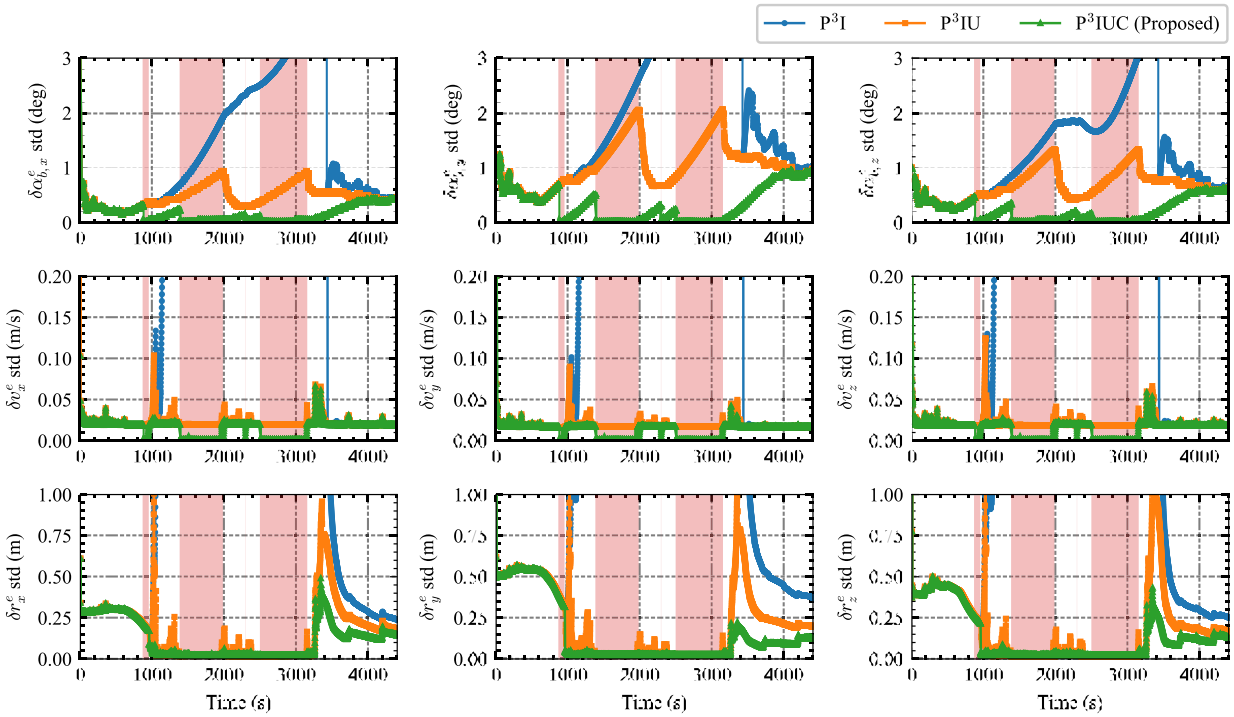


Fig. 9. Standard deviations of error state estimations (first row: attitude; second row: velocity; third row: position; first column: *x*-axis; second column: *y*-axis; third column: *z*-axis). The red background color indicates the ZUPT and yaw constraint periods.

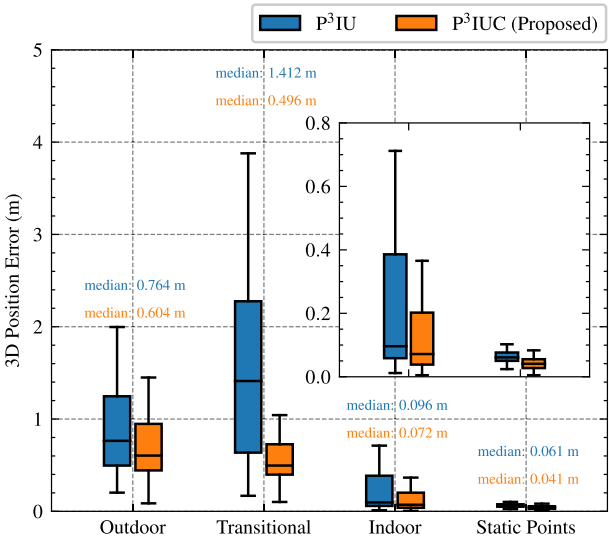


Fig. 10. Boxplots of 3-D positioning errors in different scenarios.

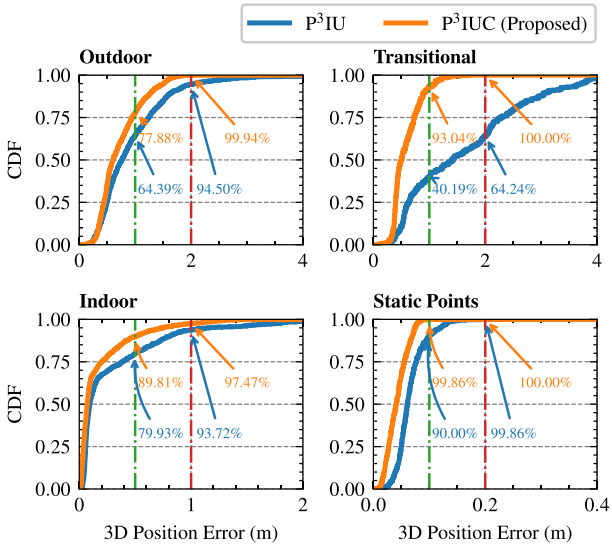


Fig. 11. CDFs of 3-D positioning errors in different scenarios.

and 100% 3-D errors less than 1 and 2 m in this area, enabling seamless positioning in indoor, outdoor, and transitional areas.

D. Area Identification and Weight Adjustment

In order to fully verify the area identification in Algorithm 1, in addition to the experiment (exp-1), a real experiment (exp-2) and two simulation experiments (simu-1 and simu-2) are carried out. The anchor positions, trajectories, and identification results of the experiments are shown in Fig. 12. Among them, the trajectory and anchor arrangement of exp-1 are quite common, and the exp-2 trajectory circles around different anchors. There are multiple outdoor areas in simu-1 and multiple indoor areas in simu-2, both of which need to be processed in blocks. The identification results of these distinct environments show that our algorithm classifies these environments into indoor, outdoor, and transitional categories. The indoor areas identified match the real indoor environments represented by the blue background. Outdoor areas are far away from buildings, and in such areas, there is often better GNSS visibility to ensure positioning. Transitional areas are located at the indoor-outdoor intersection, which is realistic because these areas are often close to buildings and trees in the real world.

After area identification, we can delete potential low-quality anchors and adjust weights. To verify the proposed weight adjustment strategy, we compare the 3-D position error

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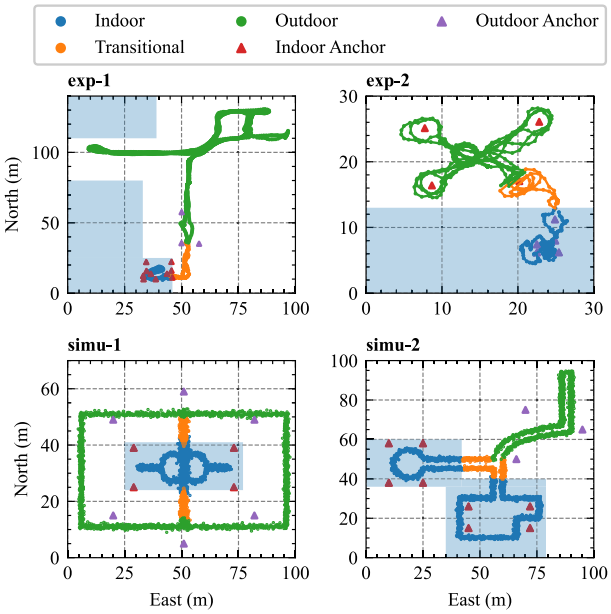


Fig. 12. Area identification results of multiple environments. The blue background color indicates indoors.

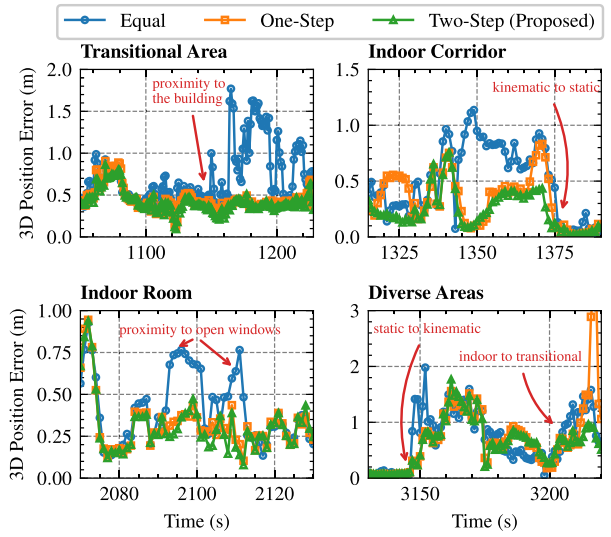


Fig. 13. 3-D position errors of different weighting methods during four challenging periods.

sequences of equal weight, one-step (segmentation weighting based on RSSI), and proposed two-step (one-step method and weight adjustment based on area identification) methods in Fig. 13. There are four challenging periods, including transitional area, indoor corridor, indoor room, and diverse area switching. It can be seen that the 3-D error of the equal weight method increases significantly when it is close to buildings and windows, which is due to NLOS, multipath and faulty signals. The one-step method introducing RSSI weighting improves the outlier solutions because RSSI reflects the effects of NLOS and multipath to some extent. However, on the one hand, the RSSIs output by hardware may not be completely accurate, and on the other hand, there may be faulty signals with gross errors that are not excluded. Such as 1325 and 3215 s, we find that the error of the one-step method even exceeds the

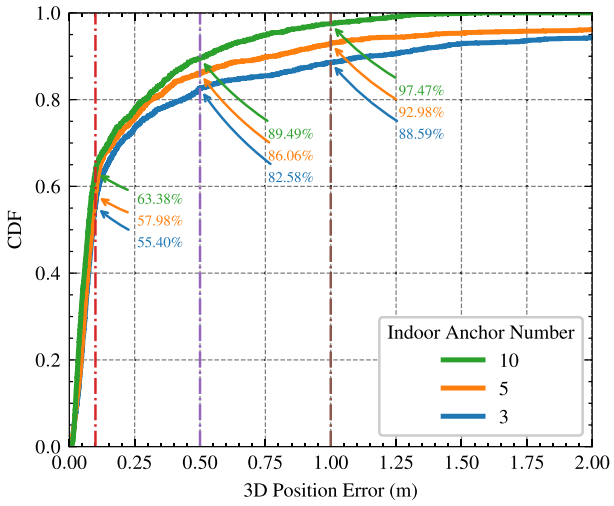


Fig. 14. Performance degradation due to indoor anchor numbers (red/purple/brown dotted lines: 0.1/0.5/1.0 m).

equal-weight method, which seriously damages the positioning continuity. The two-step method with the weight adjustment based on area identification can eliminate the faulty signals and optimize the weight model to achieve the most reliable performance during four challenging periods. In conclusion, the proposed two-step method has the most robust performance due to the synthesis of multiple factors.

E. Evaluations of Degradation and Efficiency

In this section, we explore the degradation and efficiency of the proposed method in detail. The number of anchors has a direct relation to the system cost and their geometry determines the performance of UWB positioning. In outdoor environments, GNSS is not completely lost and does not require a large number of anchors. But in indoor environments, the rangings between the tag and anchors are the only absolute information to guarantee performance in the proposed method. Hence, utilizing as few anchors as possible is also considered in our work. To this end, the number of indoor anchors is reduced from 10 to 5 and 3, and the cdfs of the 3-D positioning errors are shown in Fig. 14. For the results with 3, 5, and 10 indoor anchors, the proportions of errors less than 0.1 m are (55.40%, 57.98%, 63.38%), less than 0.5 m are (82.58%, 86.06%, 89.49%) and less than 1 m are (88.59%, 92.98%, 97.47%), respectively. The results fully demonstrate that even with only three indoor anchors, the proposed method can still achieve reliable 3-D positioning without significant degradation.

For unknown environments, the height acquisition relies on sensor measurements or filter calculations, which may be biased. The height bias given by the sensor may range from decimeters to several meters, so the potential impact of a biased height constraint on the filter is also considered in our work. The biases of 0.1, 0.5, and 1 m are added to the height constraints in this article, and the cdfs of the 3-D positioning errors are shown in Fig. 15. For the results with 1-, 0.5-, 0.1-, and 0-m biased height constraints, the proportions of errors less than 0.1 m are (42.71%, 41.31%, 42.63%, 48.68%), less

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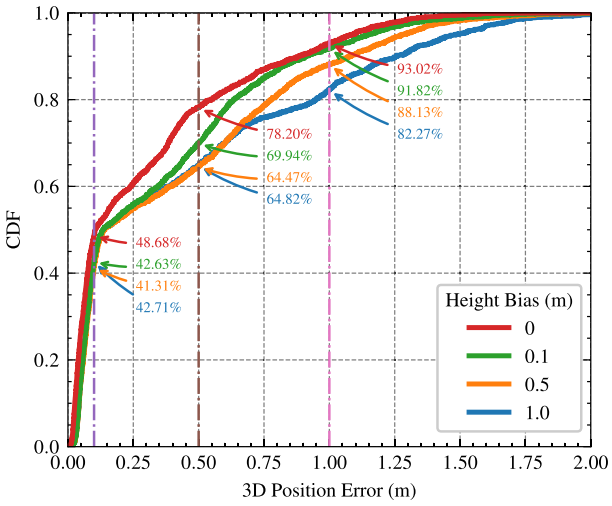


Fig. 15. Performance degradation due to height bias (purple/brown/pink dotted lines: 0.1/0.5/1.0 m).

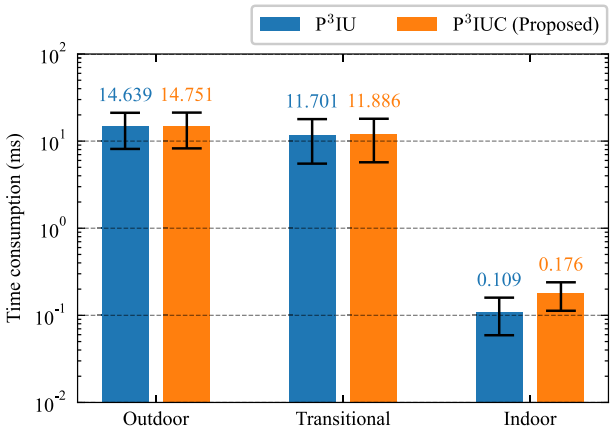


Fig. 16. Time consumption in different areas.

than 0.5 m are (64.82%, 64.47%, 69.94%, 78.20%) and less than 1 m are (82.27%, 88.13%, 91.82%, 93.02%), respectively. Overall, the proposed method in this article still does not degrade significantly even with a 1-m height constraint bias.

Then comes the evaluation of efficiency, which is the most critical for real-time applications of our method. We counted the average time consumption of processing all data collected each second in different areas, as shown in Fig. 16. In outdoor, transitional, and indoor areas, the average consumptions of the P^{3}IU method are (14.639, 11.701, and 0.109 ms), while those of the proposed method are (14.751, 11.886, and 0.176 ms). The time costs in outdoor and transitional areas are much longer than indoors because of the larger dimension of GNSS data. Regardless of the area, for all data collected in a second, the proposed system can process within 15 ms. Compared with the conventional P^{3}IU method, the additional time consumption introduced by our method is negligible.

V. Conclusion

In this article, we propose a low-cost, TC vehicle integration system that fuses heterogeneous measurements in a lightweight EKF framework, including SD-IF pseudorange and carrier-phase from a single GNSS receiver, IMU records, UWB ranging, and motion constraints for seamless indoor and outdoor positioning. In outdoor areas, the PPP technique based on SD-IF measurements from a single receiver can guarantee high-precision positioning. For transitional areas with harsh GNSS signal quality or indoor areas, UWB measurements are used as a complement or a substitute, and motion constraints are introduced as an enhancement. For UWB NLOS, multipath, and faulty signals during area switching, a two-step weighting model including segmentation weighting based on RSSI and weight adjustment based on area identification is proposed. Due to the mutual assistance between inertial measurements and heterogeneous sensors, we can achieve continuous positioning and orientation across areas indoors and outdoors.

To evaluate the performance of the proposed system, experiments were conducted on campus including open, shaded, and indoor environments. The results confirm that the proposed system equipped with a low-cost sensor suite can provide high-precision indoor-outdoor seamless positioning kinematic solutions with over 90% availability. The MAEs in the east, north, and up directions are (0.214, 0.186, and 0.243 m), and the RMSEs are (0.339, 0.291, and 0.414 m). Compared with groundtruth, the proposed multisensor TC system can significantly improve positioning performance in transitional areas and achieve smooth and seamless indoor-outdoor switching. Also, two real-world and two simulated experiments verify that the proposed UWB two-step weighting model can achieve accurate area identification for distinguishing faulty signals and adjusting the weights to ensure continuous positioning in different environments. Even with only three indoor anchors, or a biased height constraint of 1 m, there is no crashing performance degradation. For processing all data collected within a second, our method takes less than 15 ms in outdoor and transitional areas, and less than 0.18 ms in indoor areas, which is very suitable for real-time applications.

In future work, we consider fusing the proposed method with popular simultaneous localization and mapping (SLAM) techniques to achieve ubiquitous positioning, mapping, and navigation applicable to arbitrary scenarios.

Acknowledgment

The algorithm implementation is based on the GNSS+ REsearch, Application, and Teaching (GREAT) software developed by the GREAT Group, School of Geodesy and Geomatics, Wuhan University.

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